Cover letter

Dear Editor-in-Chief,

We submit the manuscript entitled “Py-CoSim: an open-source, pure-Python framework for synchronized co-simulation of EnergyPlus buildings with energy, mobility and artificial-intelligence components” for consideration as a Research Article in Building Simulation (section: Advances in Modeling and Simulation Tools). The paper presents an open-source, pure-Python co-simulation framework that turns EnergyPlus into a steppable component and couples it with energy, mobility and artificial-intelligence models on a verified common clock. We show bit-identical agreement with stand-alone EnergyPlus, parallel multi-building scalability and three case studies (grid-interactive community, LLM supervisory control with guardrails, travel-behavior-driven occupancy). The manuscript has not been published and is not under consideration elsewhere; an early prototype was posted as a preprint (SSRN 4572925), which the present work substantially extends and supersedes. All authors have approved the submission.

Sincerely, [Corresponding author name]

Py-CoSim: an open-source, pure-Python framework for synchronized co-simulation of EnergyPlus buildings with energy, mobility and artificial-intelligence components

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Author contributions (CRediT)

[Author 1]: Conceptualization, Methodology, Software, Validation, Writing – original draft. [Author 2]: Data curation (NHTS), Investigation. [Author 3]: Data curation, Writing – review & editing. — placeholders; please edit.

Declarations

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Competing interests. The authors have no competing interests to declare that are relevant to the content of this article.

Compliance with ethical standards. This study does not contain any studies with human or animal subjects performed by any of the authors. The National Household Travel Survey data used are public, de-identified data.

Code and data availability. Py-CoSim is available under the MIT license at https://github.com/lz356/Py-CoSim (version 2.0.0; [archive DOI, e.g., Zenodo, to be added]). The repository contains the scripts that reproduce all results and figures (paper/experiments).

Use of generative AI. During the preparation of this work the authors used Claude (Anthropic) to assist with software development, the design and execution of the computational experiments and the drafting of the manuscript. After using this tool, the

authors reviewed, tested and edited the software and the content as needed and take full responsibility for the content of the publication.

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